

QUIZ #2 for Section 1

This is a 30-minute quiz

Student ID:	Student Full Name:
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Write a pandas program that builds the following Series named `population` from the Python dictionary `{'Ankara': 5700000, 'Istanbul': 15900000, 'Izmir': 4400000, 'Bursa': 3100000}`. Print its values, its index, and its `dtype`. Then construct a new Series named `pop_subset` by reindexing `population` to the index `['Adana', 'Ankara', 'Bursa', 'Istanbul']`; print the resulting Series and use `pd.isnull()` to identify which cities have missing data. Next, multiply `population` by `1.05` to simulate a 5 % population growth and print the result, verifying that the index labels are preserved. Finally, construct a DataFrame named `city_df` from the dictionary `{'city': ['Ankara', 'Istanbul', 'Izmir', 'Bursa'], 'population': [5700000, 15900000, 4400000, 3100000], 'area_km2': [25706, 5461, 12012, 10882]}`, retrieve the `'population'` column as a Series using both dict-like notation and attribute access, access the row for `'Istanbul'` using `loc`, and add a new column named `'density'` computed as `population / area_km2`, printing the complete DataFrame at each step.